Concern Resolution Process for In-house Manufacturing

QC Story

QC STORY PROJECT

Project Title:

"To eliminate the Defect of I shaft steering column insertion hard from Marazzo Model."

Mahindra and Mahindra Ltd,NASIK_ Plant, _SCO/XYLO TCF.

	Plan	Actual
Start Date	10.06.2019	10.6.2019
Completion Date	30.10.2019	30.10.2019

Current Status: **Completed**

Phase II – QC Story

- Team Information
- Step1: Select the Theme
- Step 2: Justify the choice
- Step 3: Understand the current situation
- Step 4: Select Targets
- Step 5: Analysis (Level 1 Analysis)
- Step 6: Implement corrective measures
- Step 7: Confirm the Effects
- Step 8: Standardize
- Step 9: Summaries & Plan future actions
- Guidelines to user

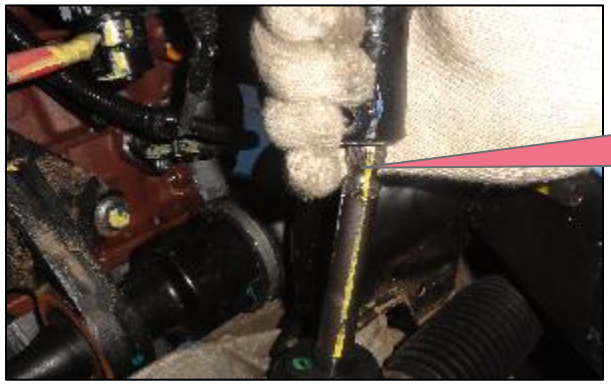
Cross Functional Team

Sr. No	Description	Name	Designation	Dept	Phone
1	**Team Leader**	Hemant Khare	L8-O	STCF	9766874268
2	**Mentor**	Rahul Narkar	DGM	STCF	9881080299
3	**Mentor**	Taj Shaikh	L6-M	STCF	9850980299
4	**Member 1**	Sayali More	L8-O	Process	7262090356
5	**Member 2**	Nilesh Deore	L8-O	STCF	8888821705
6	**Member 3**	Santosh Jadhav	L7-0	MQA	9423309400
7	**Member 4**	Jitendra Bhambre	L7-0	Maint.	8975856767

Step1. Select Subject

Brief Description : I-shaft column insertion hard in Marazzo vehicle at final line body drop under pit stage is repeat V1 concern & average 2-3 vehicles reported per day.

- What : I-Shaft Column insertion hard in Marazzo vehicle.
- Where : At final assembly line, body drop stage.
- When : Since Jun 2019.
- Who : By bolt tightening by associate at Final line assembly.
- Why : Repeat issue in CAI,V1 severity concern may leads to body panel rusty.
- How : Avg. 2 vehicles per day.



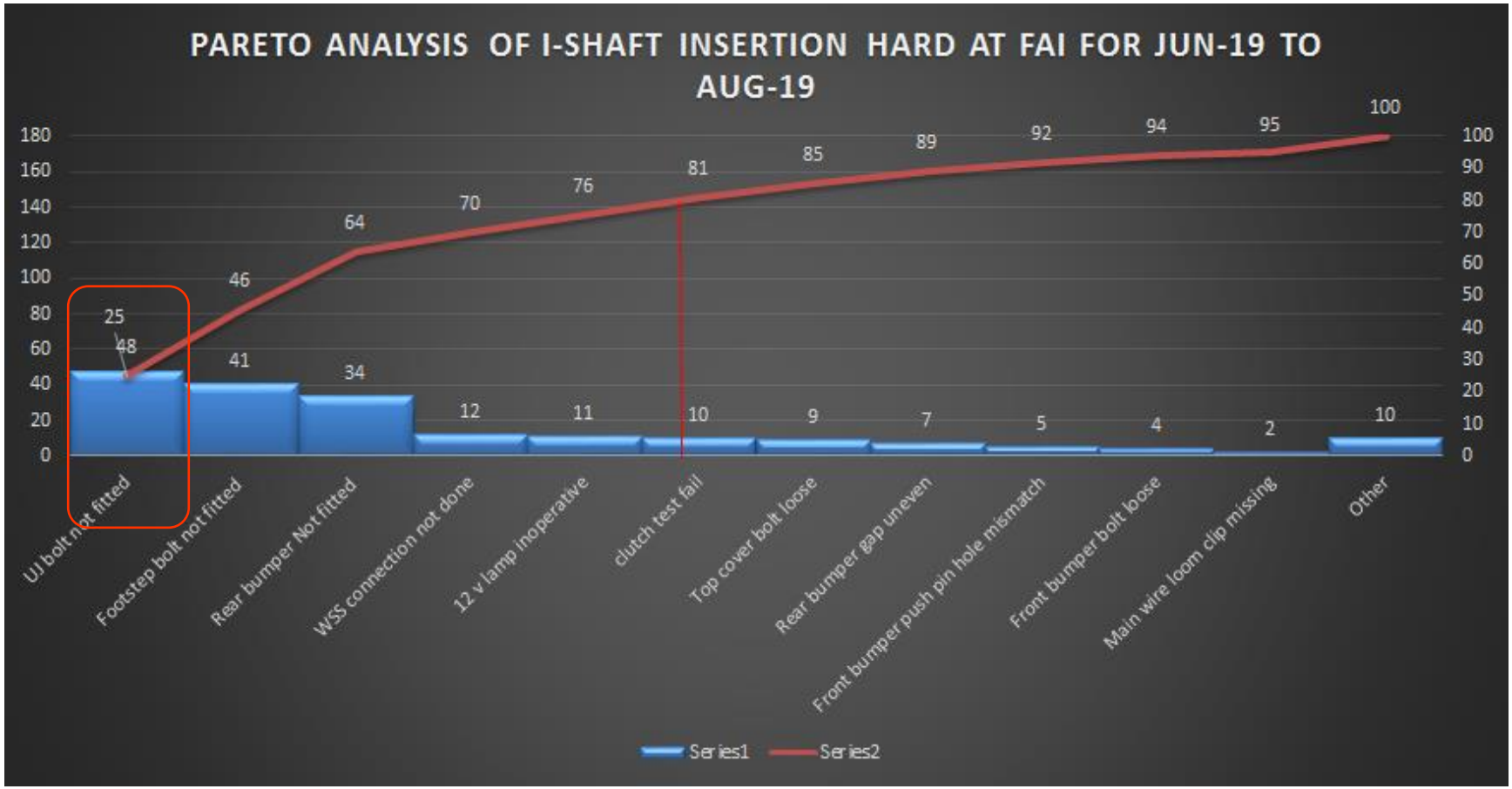
I shaft column insertion hard

2. Justify the choice (reasons)

Crusade 2.0

Departmental Objective :-
BIQ – DPR Improvement
- FRC / FBO Improvement

2. Justify the choice



Inference:

As per above Pareto Graph UJ Bolt not fitted due to insertion hard is top concern at FAI

2. Justify the choice (reasons)

Importance & Severity of Concern:

- I shaft Column insertion hard V1 + concern.
- Major contributor in FRC on FAI
- Avg. 16 cases reported in month
- Skill operator required for rework.
- Rework activity is critical in offline activity
- Repair cost is very higher RS 29,032 per year.

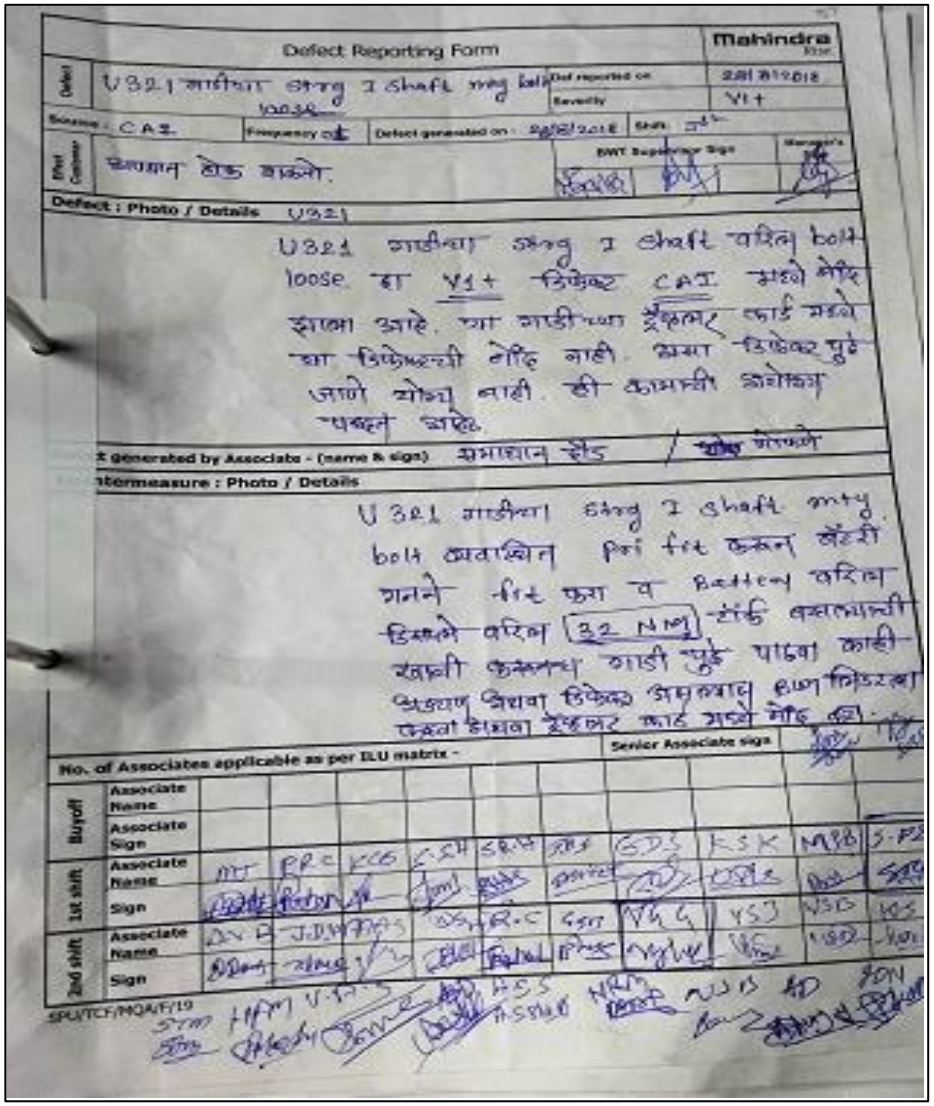
:

Rework Cost Estimation		
Description	Units	
Rewok Time	Min	15
Avg Concerns/Day	Nos	2
Total Required Rework Time per day	Min	30
Effective Working time	Min	465
Operator Require Per Day	Nos	0.064516
Approximate Cost Per Day	Rs	1500
Operator Cost/ Concern	Rs	96.77419
Avg no of working days	Days	25
Operator Cost Per Month	Rs	2419.355
Operator Cost /Year	Rs	29,032

2.b Immediate Containment Action

1. Vehicle reworked at FAI rework area for the defect UJ bolt not fitted (I shaft insertion and up down movement hard).

2. Defect communicated to the stage operator through DRF.



Defect Reporting Form

Mahindra

Defect: U321 गाडीचा Stag I Shaft मध्ये bolt loose

Defect reported on: 28/8/2018

Severity: V1+

Source: CAI

Frequency of: 1

Defect generated on: 28/8/2018

Shift: 3rd

Effect Customer: समाधान होऊ शकतो.

BWT Supervisor Sign

Manager's

Defect : Photo / Details U321

U321 गाडीचा Stag I Shaft वरील bolt loose हा V1+ डिफेक्ट CAI मध्ये नोंद झाला आहे. या गाडीच्या ट्रॅकर कार्ड मध्ये या डिफेक्टची नोंद नाही. असा डिफेक्ट पुढे जाणे योग्य नाही. ही कामाची अयोग्य पकडले आहे.

Defect generated by Associate - (name & sign) समाधान शेंडे / [illegible]

Intermeasure : Photo / Details

U321 गाडीचा Stag I Shaft मध्ये bolt अवाढ्यित fit करून बॅटरी गनने fit करा व Battery वरील टिकले वरील 32 NM टॉर्क बसलाची खात्री करून गाडी पुढे पाठवा काही अडचण असेल डिफेक्ट असल्यास BWT सिस्टम ट्रॅकर कार्ड मध्ये नोंद करा.

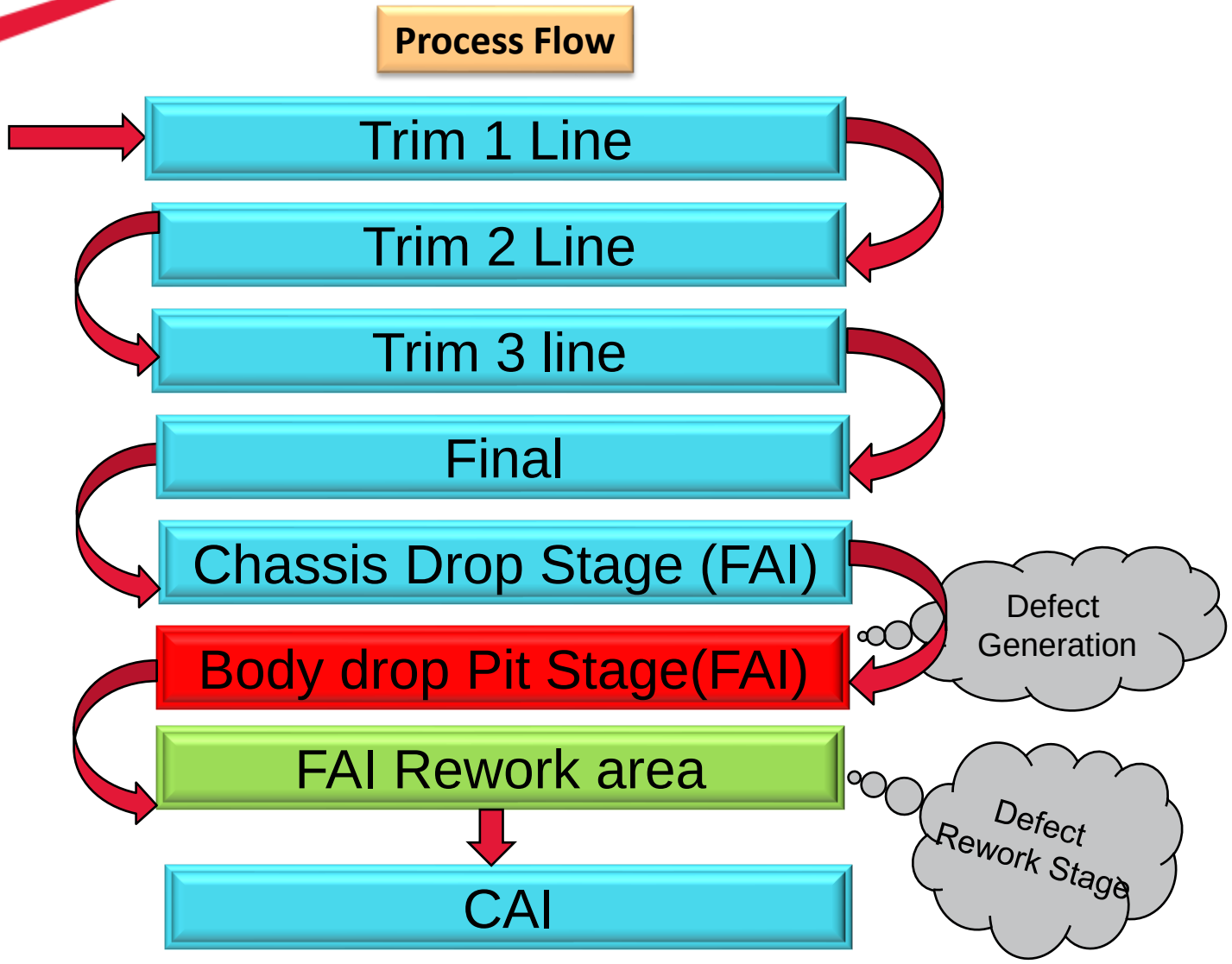
No. of Associates applicable as per DLU matrix -

Senior Associate sign

Buyoff	Associate Name										
	Associate Sign										
1st shift	Associate Name	MT	PRC	KCG	C.SH	S.R.M	[illegible]	G.D.S	K.S.K	M.P.B	S.P.Z
	Sign	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2nd shift	Associate Name	D.V.R	J.D.M	[illegible]	D.S.H	R.C	S.S.T	V.H.G	Y.S.J	V.S.D	[illegible]
	Sign	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

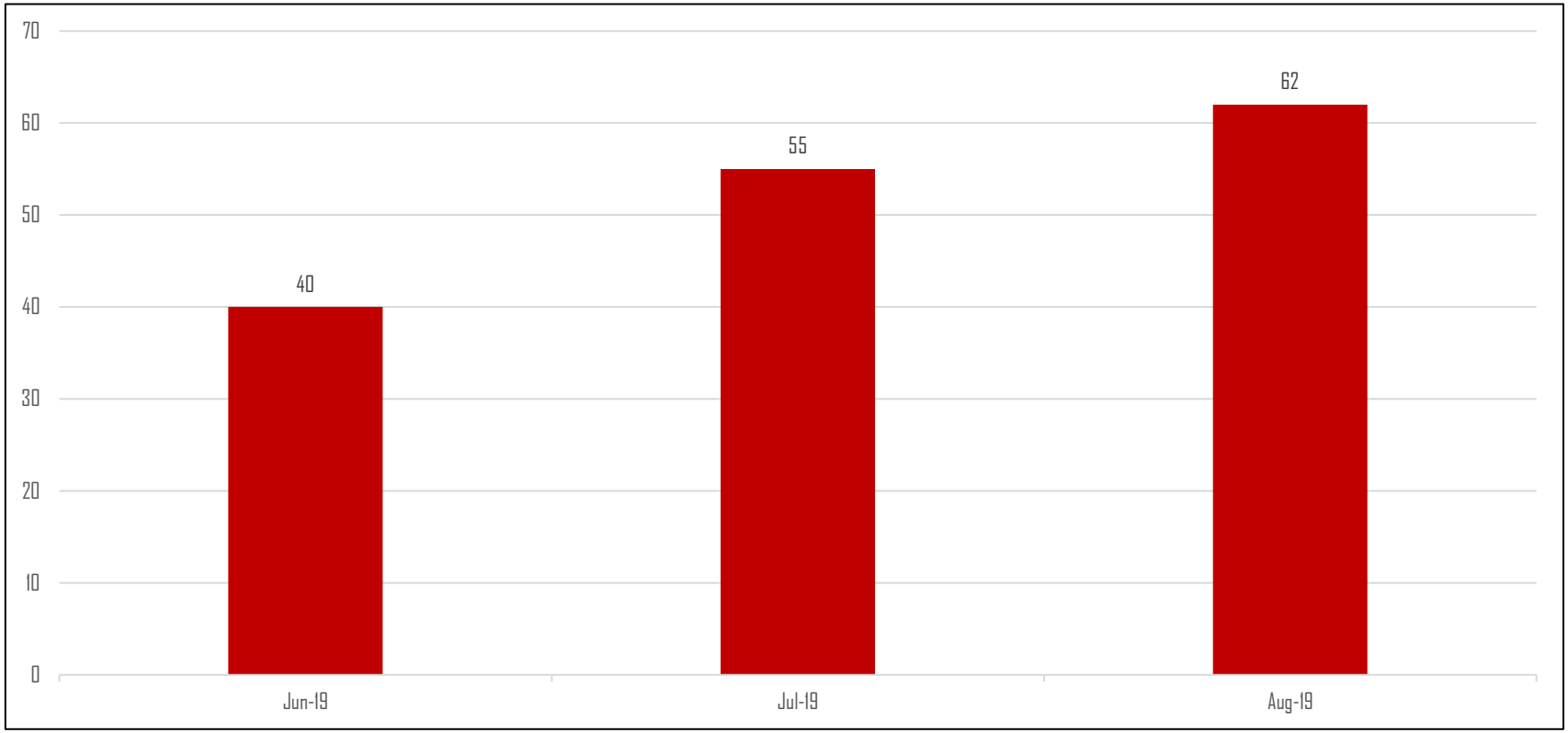
SPU/TCF/MQA/F/19

3. Understand the Current Situation



3. Understand the Current Situation

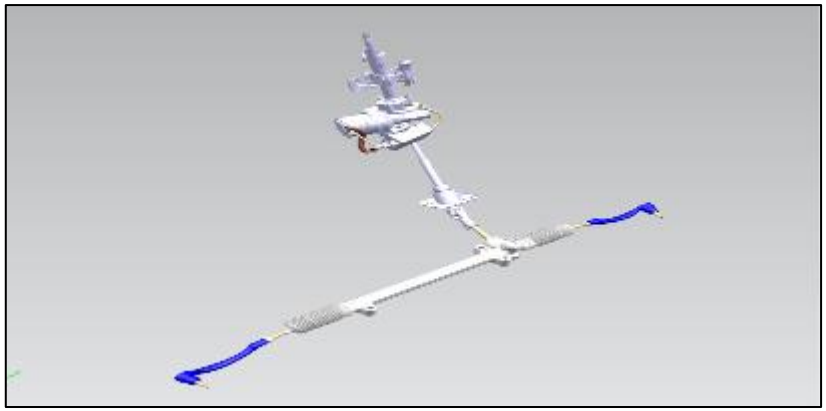
Month wise I shaft column insertion hard concern Trend at final line for June-19 to Aug-19

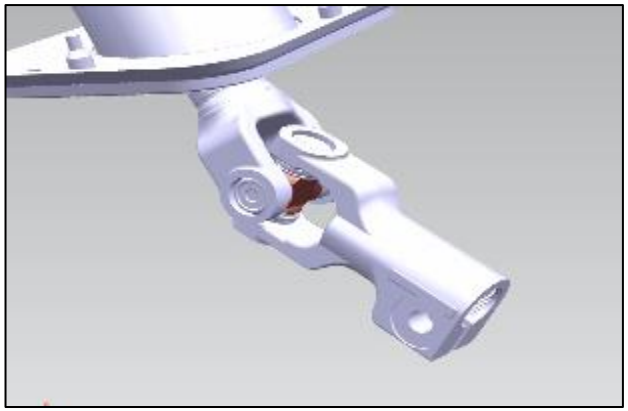


Inference – Month on month defect trend is increasing

3. Understand the Current Situation

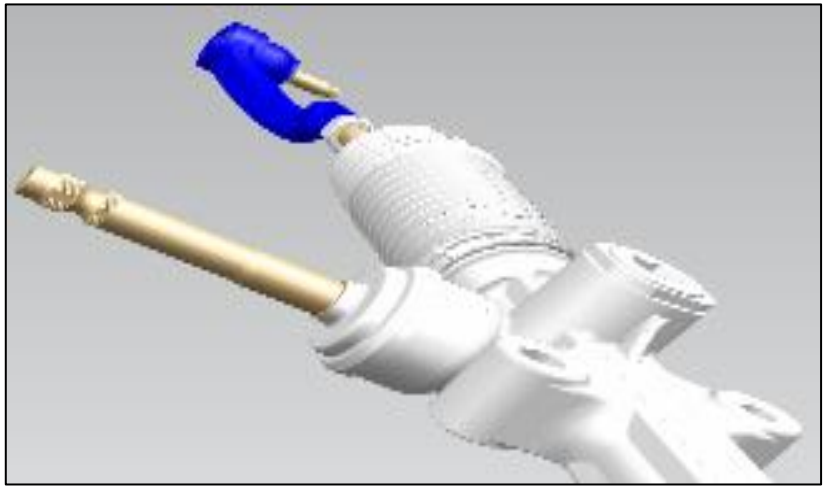
I shaft fitment Process





I shaft Steering column Assembly

Steering column



Power Steering Rack

3. Understand the Current Situation

Mahindra **4M Analysis (Part 1)**
(Conformity Analysis - 'L0')

PU/ Area : Scorpio/Xylo TCF
Cell / BWT / Line: BWT 10

Requested By- MQA | Date -14.10.2019 | Source -CAI | 4 M Analysis Part 1 Sr.No :

Problem / Concern Statement :Steering column insertion hard

FMEA Severity /Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3 | No of Incidences on the day (Including Yard check) : 1 | VIN / Aggre. No.- | Model :marazzo

Date of Incidence/ Assy : 11.10.2019 | Shift / Time : 2nd | Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
Vehicle reworked

Does a related 4M Analysis - Part 1 or QC story already exist ? ☐ YES ☑ NO | If Yes | 4M Analysis Part 1 Sr. No. / QC Story Sr. No

Failure mode or cause/s identified -

Problem Assigned to - BWT . :

Does a control already exist ? ☐ YES ☐ NO

Stage / Post Analysis -

Operator Name : kunal salve

Man(Human Action)

Is operator trained ?	☑ YES ☐ NO
Operator level (L,U)	L
OR Dexterity level(1...4)	3
Is operator under training ?	☐ YES ☑ NO
Is operator temporary for Stage ?	☐ YES ☑ NO
Bad rework ?	☐ YES ☑ NO
Is the specified /Tool / Fixture facility used ?	☑ YES ☐ NO
Last calibration dates	
Facility /Tool / Fixture in good condition ?	☑ YES ☐ NO
Is it a substitute facility ?	☐ YES ☑ NO
Are machine parameters known ? (Record in Analysis table)	☑ YES ☐ NO
Parameters are respected ?	☑ YES ☐ NO

Method(process)

Main steps are respected ?	☑ YES ☐ NO
Is there a related key point in SOP / SOS?	☑ YES ☐ NO
Are the reasons for key point known ?	☑ YES ☐ NO
Are operating reqmnts respected ?	☑ YES ☐ NO
Is the defect part of the Buy off / Checkman list ?	☑ YES ☐ NO
Was it detected by Inspector/ Buoff Opr ?-	☐ YES ☐ NO
In only one shift ?	☑ YES ☐ NO
In only one line ?	☑ YES ☐ NO
The problem follows the vehicle ?	☑ YES ☐ NO
The problem follows the part ?	☐ YES ☑ NO
Is it a repeat problem ?	☐ YES ☑ NO
Is the suspected parameter measurable ? (Record in Analysis table)	☐ YES ☑ NO

Other

Is there a Control Plan ? (Record in Analysis table)	☑ YES ☐ NO
Is the packaging standard defined for the part ?	☑ YES ☐ NO
Is the packaging standard respected ?	☑ YES ☐ NO
Is there any alert mechanism for failure ?	☑ YES ☐ NO
Is the specified product used ?	☑ YES ☐ NO
[redacted] Chan[redacted] YE[redacted] ☑ NO	
, SOP [redacted] respected ?	☑ YES ☐ NO
Is the shelf life / expiry date respected ?	☑ YES ☐ NO
Is the part as per FIFO?	☑ YES ☐ NO

Machine | **Product Failure** | **Material/ Consumable**

Note : 1. If both fields (Yes/NO) are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusionon Gemba (Attach Sketch/ Photograph if necessary) -
Vehicle reported for I shaft insertion hard
History - This vehicle was rolled down on 11.10.2019 in 2nd shift shift
Process flow - I shaft column is fited at body drop stage
Detection -I shaft insertion and fitment is checked at under pit rolldown by yellow marking .
Traveler card - No remark related to ishaft fitment in traveller card.
Packaging - NA
Defect history - NIL
Failed part :I shaft not fitted
During analysis it is observe i shaft not match with splines due to mis align of marking on steering column and power steering rack.

Sno	Product / Process / Machine / Parameter	Spec.	Actual	Remark

Is 4M Analysis Part 2 required ? ☐ YES ☑ NO

Is the substitute transport mode used ? ☐ YES ☑ NO **Is the packaging defined ?** ☑ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
Awarness given to respective associates through DRF	Hemant	11.10.19	Closed

2. Permanent Action -	Resp	Date	Status
1) handy fixture to be provided to match splinge of steering column with power steering rack.	Hemant	27.10.19	Close

Analysed by Name/s - Taj shaikh
Date - 26.6.19

Choice : ☐ WSS ☑ QC Story

4M analysis Good and Not so good observations

MAN	Main step are respected by associate.
Method	Operator aware of key point and check present at buyoff
Others	Control plan adhered on stage
Machine	Tools used are calibrated
Product failure	Problem follows Vehicle
Material	Handling and shelf life of part adhered

No Quick Win

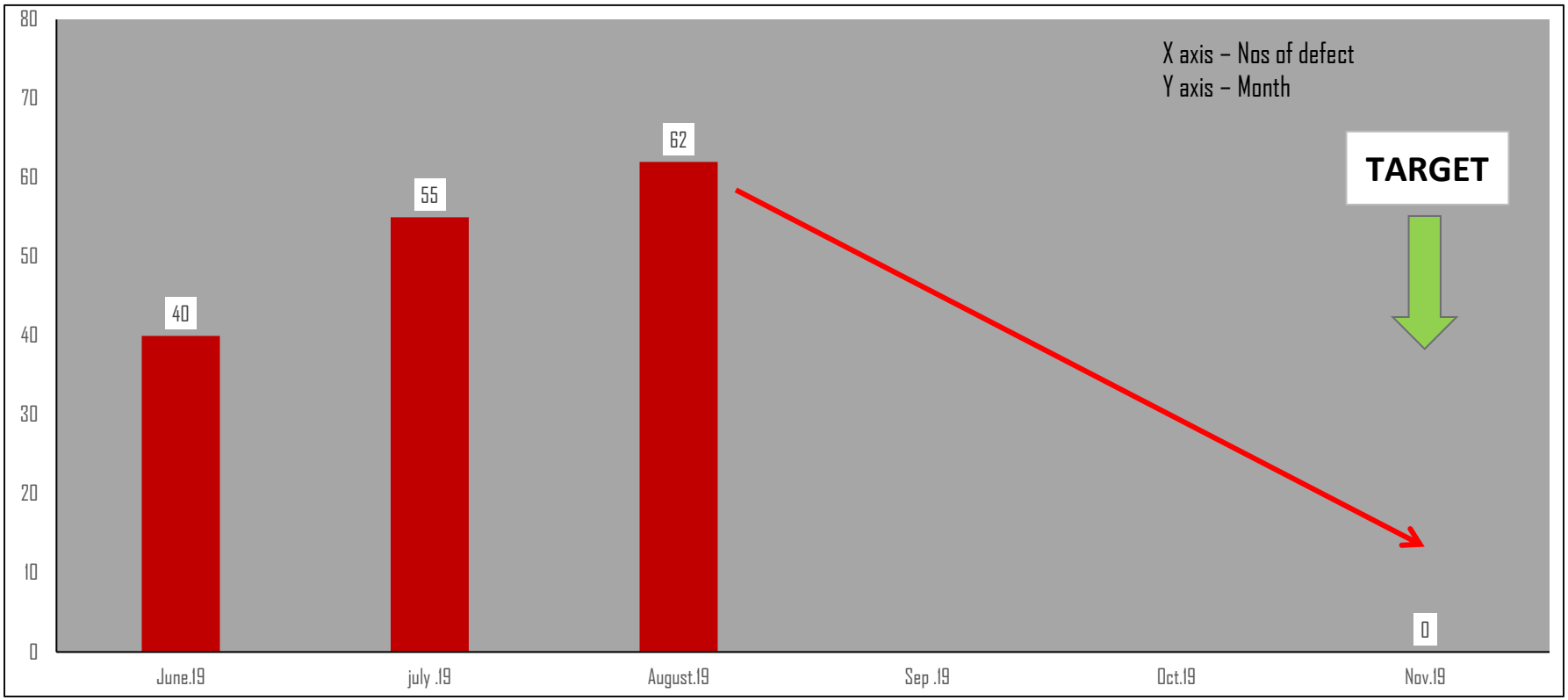
3. Understand the Current Situation

4M Analysis Observation

- Concern Reported in Pit at Roll down stage.
- All L3 Operators Working on Stage.
- Concern Reported in Marazzo all models .
- Concern reported as UJ bolt Not fitted (I shaft column insertion hard).
- Rework is done at the rework area at pit area - not possible in tact time on stage .
- No access for rework on line.
- Rework Takes more than 15-20 min time.

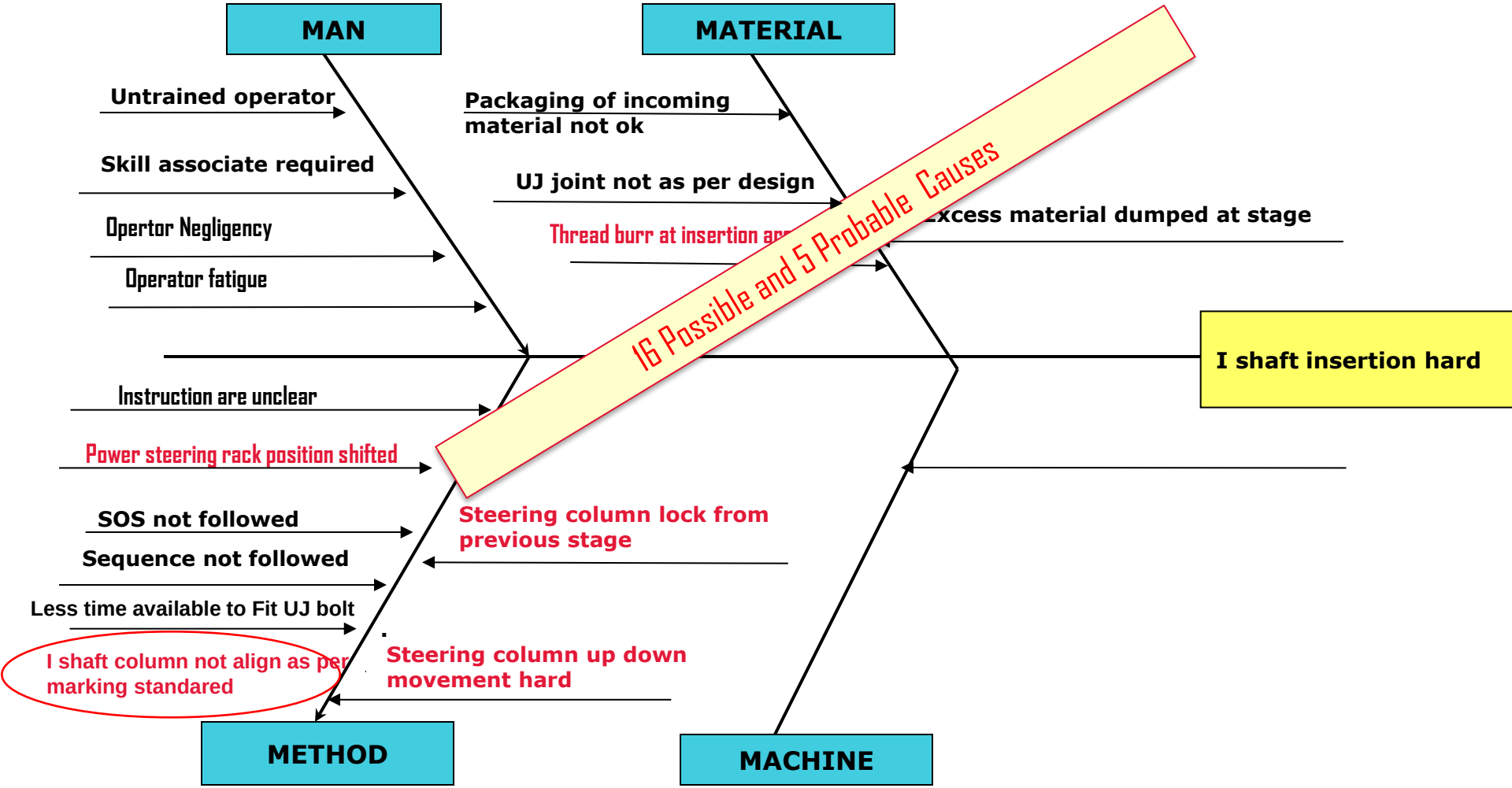
4. Select Target

Present Level:	Average 52 Defects/Month
Target :	Elimination of the defect I shaft column insertion hard .



5. Analysis

List of Possible Causes



5. Analysis

5.1 List of Possible causes: Brainstorming

Sr no	Possible cause	Nilesh	Sachin Bhambra	Santosh jadhav	Jitendra Bhambra	Dilip junnare	Total
1	Untrained Operator	1	1	1	1	3	7
2	High skill need to fit the bracket	1	1	1	1	3	7
3	Operator Negligency	1	3	1	3	3	11
4	Operator fatigue	1	1	1	1	1	5
5	Packaging of incoming material not ok	1	1	1	1	1	5
6	UJ joint not as per design	3	3	3	3	3	15
7	Thread burr at insertion area	9	9	9	9	9	45
8	Excess material dumped at stage	3	3	3	3	3	15
9	Instruction are unclear	1	1	1	1	1	5
10	Power steering rack position shifted	9	9	9	9	9	45
11	SOS not followed	3	3	3	3	3	15
12	Sequence not followed	3	3	3	3	3	15
13	Less time available to fit UJ bolt	3	3	3	3	1	13
14	I shaft column not align as per marking standared	9	9	9	9	3	39
15	Steering column lock from previous stage	9	9	9	9	9	45
16	Steering column up down movement hard	9	9	9	9	3	39

Selection Criteria: 0-No impact, 1- less impact , 3-High impact , 9- Significant impact.
Threshold Value for selecting probable cause is 39.

Step 5. Analysis

5.1 List of Probable causes: Brainstorming

List of causes not considered as probable causes and reasons for not considering

Sr nos	Possible cause	Reasons
1	Untrained Operator	Trained operator deploy on stage
2	High skill need to fit the bracket	L3 Dexterity level deploy on stage
3	Operator Negligency	No operator negligency observe
4	Operator fatigue	No Operator fatigue observe for fitment
5	Packaging of incoming material not ok	Packaging as per packaging standared
6	UJ joint not as per design	Part found as per design
8	Excess material dumped at stage	Two bin system available at stage
9	Instruction are unclear	Instruction clear mentioned in SOS
11	SOS not followed	SOS followed by associate
12	Sequence not followed	Sequence followed by associate
13	Less time available to fit UJ bolt	Sufficient time allocate to complete work

5. Analysis

5.1 List of Probable causes: Brainstorming

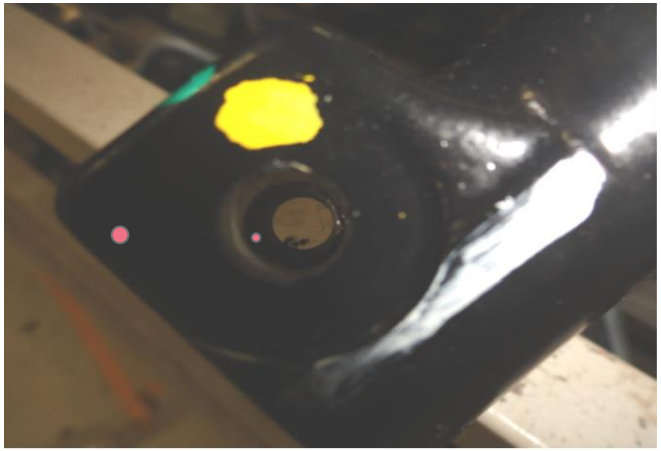
Sr nos	Possible cause
7	Thread burr at insertion area
10	Power steering rack position shifted
14	I shaft column not align as per marking standared
15	Steering column lock from previous stage
16	Steering column up down movement hard

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
7	Thread burr at insertion area	Visual	25 Nos I shaft Column check burr found at insertion area .	Hypothesis valid

Thread found Not ok



Sr no	Thread check
1	found ok
2	found ok
3	found ok
4	found ok
5	found ok
6	found ok
7	found not ok
8	found ok
9	found ok
10	found ok
11	found ok
12	found ok
13	found ok
14	found not ok
15	found ok
16	found ok
17	found ok
18	found ok
19	found ok
20	found ok
21	found ok
22	found ok
23	found ok
24	found ok
25	found ok

Concern	No Burr	Burr
Vehicle Checked		

25 no of part check for burr and out of it 2 part found burr

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
10	Power steering rack position shifted	1) Visual observations	CFT member observe that stage for 2 hrs. Observations are as below… 1) Power steering rack position found shifted at time of wheel fitment .	**Hypothesis valid**

Power steering rack position disturbed

20 Nos of part check for power steering position and out of it 4 part found position shifted.

Concern	Position found ok	Position disturbed
Vehicle Checked		

Sr no	Model	position
1	M2	ok
2	M4	ok
3	M6	Not ok
4	M2	ok
5	M8	ok
6	M4	ok
7	M6	ok
8	M4	Not ok
9	M8	ok
10	M2	ok
11	M6	ok
12	M4	ok
13	M8	ok
14	M2	ok
15	M4	Not ok
16	M8	ok
17	M6	ok
18	M8	ok
19	M6	ok
20	M4	Not ok

At body drop under pit it is observe that power steering rack position found disturbed.

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
14	I shaft column not align as per marking standard	Visual match marking	10 Nos of vehicle check for match marking on column and it is found I shaft column align as per marking standard..	Hypothesis Invalid

I shaft column insertion as per match marking on power steering rack and UJ column

Concern	Insert with Match marking	Insertion without match marking
Vehicle Checked	10	

10 Nos of vehicle check and it is found that I shaft align as per marking std.

Sr nos	Model	Position
1	M2	Found ok
2	M4	Found ok
3	M6	found ok
4	M2	found ok
5	M4	found ok
6	M8	found ok
7	M6	found ok
8	M4	found ok
9	M8	Found ok
10	M2	found ok

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
15	Steering column lock from previous stage	Visual	It is observe that steering column found in lock position from previous stage	Hypothesis valid

Steering column found in lock position

Sr no	Steering column position
1	found not ok
2	found ok
3	found ok
4	found ok
5	found ok
6	found ok
7	found not ok
8	found ok
9	found ok
10	found ok

Concern	Unlock key position	Lock position
Vehicle Checked	|||||	||

10 Nos of vehicle check and it is observe that 2 vehicle Steering column found in lock position .

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
16	Steering column up down movement hard	Inserting steering column into power steering rack	During observation it is found that UJ column up down movement found hard .	Hypothesis valid

Steering column up down movement found hard .

UJ column found hard during insertion into power steering rack

Sr nos	Model	Uj column insertion
1	M2	Found ok
2	M4	Found ok
3	M6	found Nok
4	M2	found ok
5	M4	found ok
6	M8	found ok
7	M6	found Nok
8	M4	found ok
9	M8	Found ok
10	M2	found ok

Concern	Easy Up down movement	Up down insertion hard
Vehicle Checked	|||||	||

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing method	Testing and Observations	Conclusion
1	Thread burr at insertion area	Thread gauge	25 Nos I shaft Column check burr found at insertion area .	Hypothesis valid
2	Power steering rack position shifted	Visual observations	1 member observe that stage for 2 hrs. Observations are as below... 1 Power steering rack position found shifted at time of wheel fitment .	Hypothesis valid
3	Steering column lock from previous stage	Vi	It is observe that steering column found in lock position from previous stage.	Hypothesis valid
4	Steering column up movement hard	Up down movement of steering column	During observation it is found that UJ column up down movement found hard .	Hypothesis valid

4 Valid Causes

5.3 Identify the root causes:

WHY - WHY Analysis

No	Valid Probable Cause	WHY	WHY	WHY	ACTION / COUNTERMEASURE
1	Thread burr at insertion area	Burr coming on parts received from supplier	Skipped from Supplier checking	100% parts not checked at supplier	Frequency to be change
2	Power steering rack position shifted	During tyre fitment on chassis it position get disturbed	No provision available for holding wheel during fitment	Process limitation	Provide the facility
3	Steering column lock from previous stage	Steering gets lock at trim 3	During other part fitment it position gets disturbed	This failure mode not aware to trim associates	Give awareness & add check point.
4	Steering column up down movement hard	Steering column lower end stick inside rubber grommet	Air getting trapped inside rubber grommet while assembly on dash panel	Design inadequate	Change the design

6. Implement the corrective measures

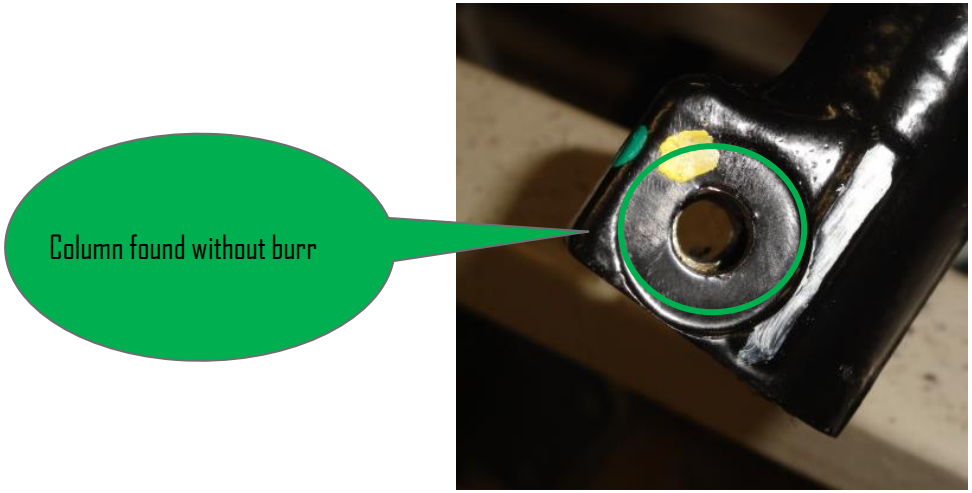
6.1 Actions Decided

No	Root Cause	Action plan	Target date	Responsibility	Status
1	100% parts not checked at supplier	Frequency to be change	20.7.19	Satish Pagar	Closed
2	Process limitation	Provide the facility	15.7.19	Hemant Khare	Closed
3	This failure mode not aware to trim associates	Give awareness & add check point.	20.7.19	Tarit Nandi	Closed
4	Design inadequate	Change the design	12.9.19	Santosh Jadhav	Closed

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
1	100% parts not checked at supplier	100 % checking done by gauge qualifying at vendor end	10.7.19	Satish Pagar	Close

Sr no	Trial taken
1	found ok
2	found ok
3	found ok
4	found ok
5	found ok
6	found ok
7	found ok
8	found ok
9	found ok
10	found ok



6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
2	Process limitation - Power steering rack position gets shifted	Provision made for required power steering rack position (Handy gauge provided for positioning of power steering rack)	15.7.19	Hemant Khare	Close

After centring by using handy fixture power steering column position is corrected .

1) During analysis it was observe that, rack position is shifted and it is easily visible as yellow marking on rack position found disturbed.
2) At console stage for Scorpio vehicle we have already using the tool to match marking over I shaft position w.r.t rack position. .

Process Change Requisition Note PCRN

DISPLAY PCRN

PCR....:* PCRN/A003/STCF/20/14553 | Use of tool for I shaft position setting | Planned closing date:* 27.10.2019 | Actual closing date:

A.Form

PU / Area:* STCF | Cell:* FAI | Date: 25.09.2019

Models Affected: Marazzo

Opn No./Mc No.: | Opn Description: Use of tool for I shaft position setting

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
- [ ] Production from tooling and equipment transferred to a different plant location or from additional plant location Production from New or Modified

Use of tool for I shaft position setting

Estimated Cost Impact

Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

- [ ] Material
- [ ] Manpower
- [ ] Conversion

E.Requesting PU/ Area / Shop

Upload Documents(Attachments limit is 1mb/PCRN no.)

A.

Is PCRN already Implemented:* ○ Yes ◉ No | Implementation Date:*

Justification:

MPPAP to be Followed:* ◉ Yes ○ No | SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

i.Supply Module ○ Yes ◉ No | ii.Maintenance Department ◉ Yes ○ No | iii.Quality Assurance ◉ Yes ○ No

Approval Status

6. Implement the corrective measures

No	Root Cause	Action plan	Target date	Responsibility	Status
3	This failure mode not aware to trim associates Steering column lock from previous stage	Awareness given at trim about unlocking of steering column	20.7.19	Tarit Nandi	Close

TRIM-3 BUY-OFF - CHECK POINTS MARAZZO

Rev no:-50 | 03.12.2019

S.N.	SIDE	SC CC	Description	What to Check	Severity	How to Check	HOW to MARK	why to check
1		SC CC	Key position	key position should be in unlock side	V2	Visually		Steering column and rack insertion assembly not posible at body drop stage.

IMS/SPU/TCF/F/04

REV NO			Date	Revision detail	Reason
1			10.03.2019	Marking standard review for G Nova C Concern Black primer stain in Lift gate top side	RFI concern
2			14.03.2019	Marking standard review for G Nova C Concern Mis alligned RH Frt door outer waist seal w.r.t.RR door outer waist seal	G Nova C
3			05.04.2019	Check sheet revised for RFI Concern Rear fix glass run seal fold top side. (Point no: 30).	RFI concern
4			12.04.2019	Check sheet revised for RFI Concern Not locked Lift Gate Garnish Rh Side (Point no: 34).	RFI concern
5			18.04.2019	Check sheet revised for RFI Concern O sticker missing at front Door. (Point no: 10).	RFI concern
6			30.5.2019	Check sheet revised for RFI Concern Captain seat back rest rexin torn	RFI concern
7			13. 6.2019	Cheeck sheet revised for RFI Concern of Tail door noise. (Point no : 32).	RFI concern
8			27.8.2019	Checksheet revised for GNOVAC defect- ORVM outer seal torn	G Nova C
9			27.8.2019	Check point for Steering coloum unlock position	FAI Concern
10			10.10.2019	Checksheet revised for GNOVAC defect- Seal lip fold Rh Rear. (Point no: 26).	G Nova C
11			11.11.2019	Checksheet revised for RFI Defect- Cross fit AC Knob. (Point no: 22).	RFI concern
12			03.12.2019	Checksheet revised for G NOVA CDefect- Not properly secured wire loom at combi shroud bottom side. (Point no: 18).	G NOVA C concern

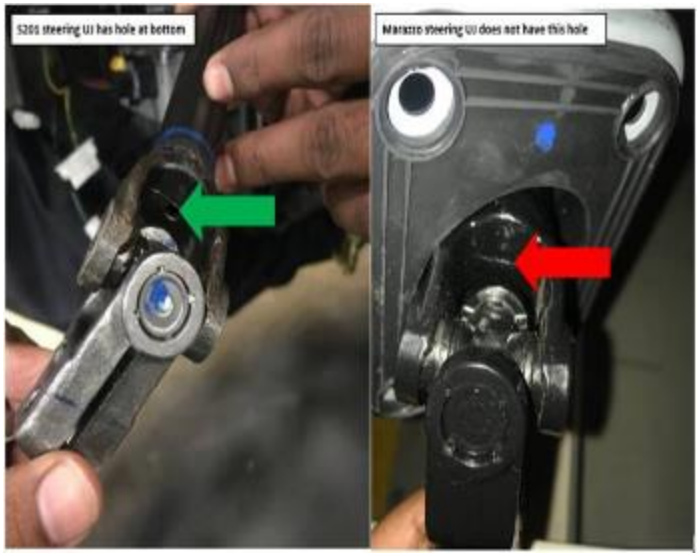
Correct key position from previous stage

Check point added at previous stage for correct key position.

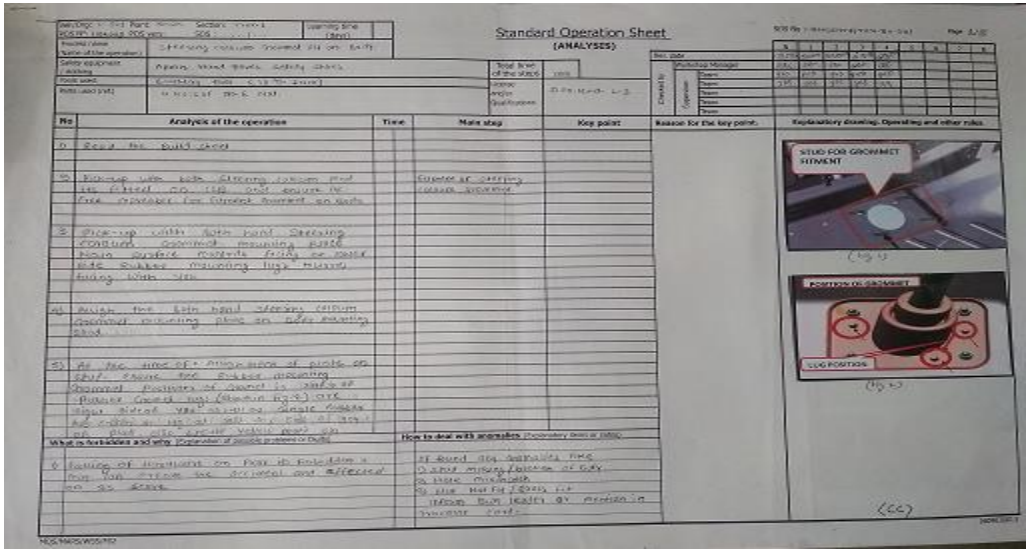
6. Implement the corrective measures

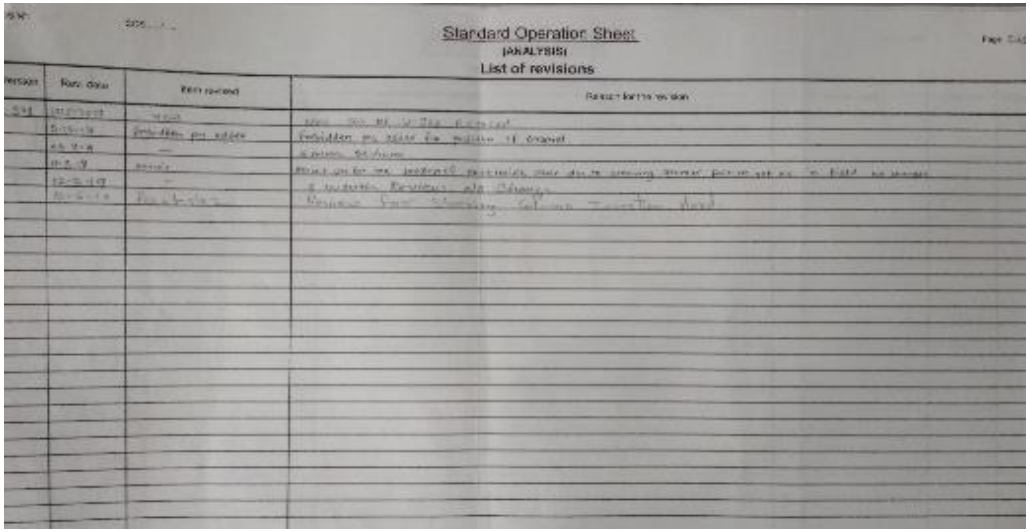
No	Root Cause	Action plan	Target date	Responsibility	Status
4	Steering column up down movement hard	Up down movement checking activity added at trim end.	10.7.19 .	Santosh Jadhav	Closed

Check point at trim end for up down movement .



Sos updated


Standard Operation Sheet
(ANALYSES)
No
Analysis of the operation
Time
Main step
Key point
Reason for the key point
STUD FOR GROMMET FITMENT
POSITION OF GROMMET
LOG POSITION


Standard Operation Sheet
(ANALYSIS)
List of revisions
Reason for the revision

7. Check & confirm the results & effects

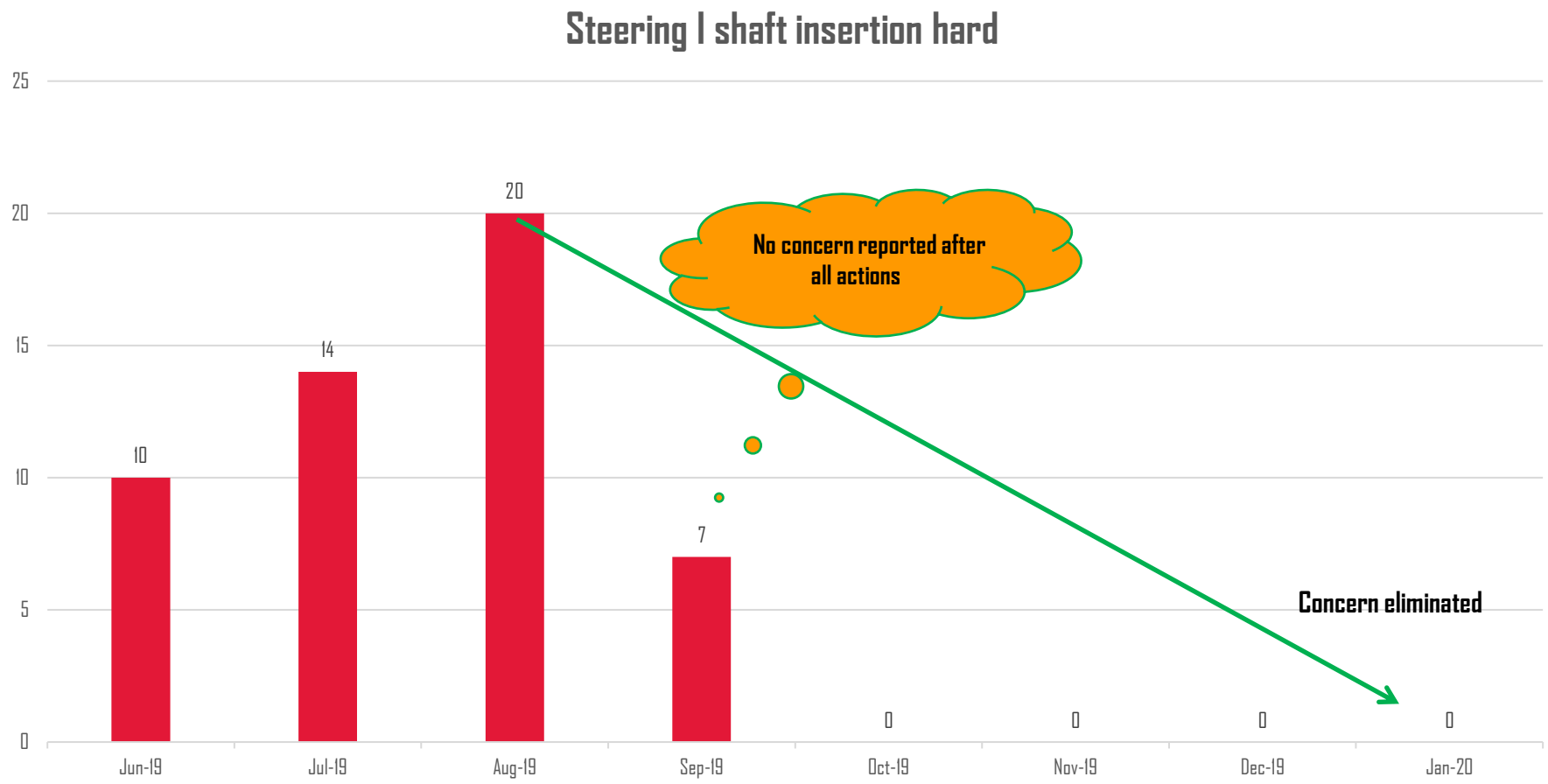
7.1 Verify Results:

S N	Action Taken	Testing Method	Results Before Action	Results After Action
1	100 % checking to be done by gauge qualifying at vendor end.	Thread gauge	25 Nos I shaft Column check burr found at insertion area	Pre fitment of bolt found ok and no burr observe during fitment.
2	Handy gauge to be provided for positioning at chassis drop stage	Visual observations	CFT member observe the stage for 2 hrs. Observations are as below... 1.In every vehicle associate facing I shaft column insertion hard .	I shaft insertion found ok by providing handy fixture for positioning of power steering rack
3	Awareness to be given at trim line about unlocking of steering column .	Visual inspection	UJ column up down and rotation not possible.	UJ column Up down and rotation found ok .
4	Steering column up down movement hard.	Up down movement of UJ column	UJ column and power steering rack insertion not possible .	UJ column get easily inserted into power steering rack.

7. Check & confirm the results & effects

7.1 Verify Results: Defect Trend

X Axis- Month
Y Axis: Defect trend history



Inference - Concern is eliminated after all 3 actions implemented.

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	100 % checking done by gauge qualifying at vendor end .	20.7.19	Satish Pagar	Closed
2	Handy gauge provided for positioning at chassis drop stage.	15.7.19	Hemant Khare	Closed
3	Awareness given at trim about unlocking of steering column.	20.7.19	Tarit Nandi	Closed
4	Up down movement checking activity added at trim end.	12.9.19	Santosh Jadhav	Closed
5	Marking to be ensure at chassis centering stage .	15.7.19	Hemant khare	Closed

7. Check & confirm the results & effects

7.2 Full Scale Implementation plan:

S N	Action Taken	Target date	Responsibility	Status
1	PCRN	27.10.2019	Sayali more	Closed
2	SOS Updation.	15.7.19	Hemant Khare	Closed
3	PFMEA updation.	27.10.19	Suhas Aher	Closed

Check & confirm the results & effects (Cont..)

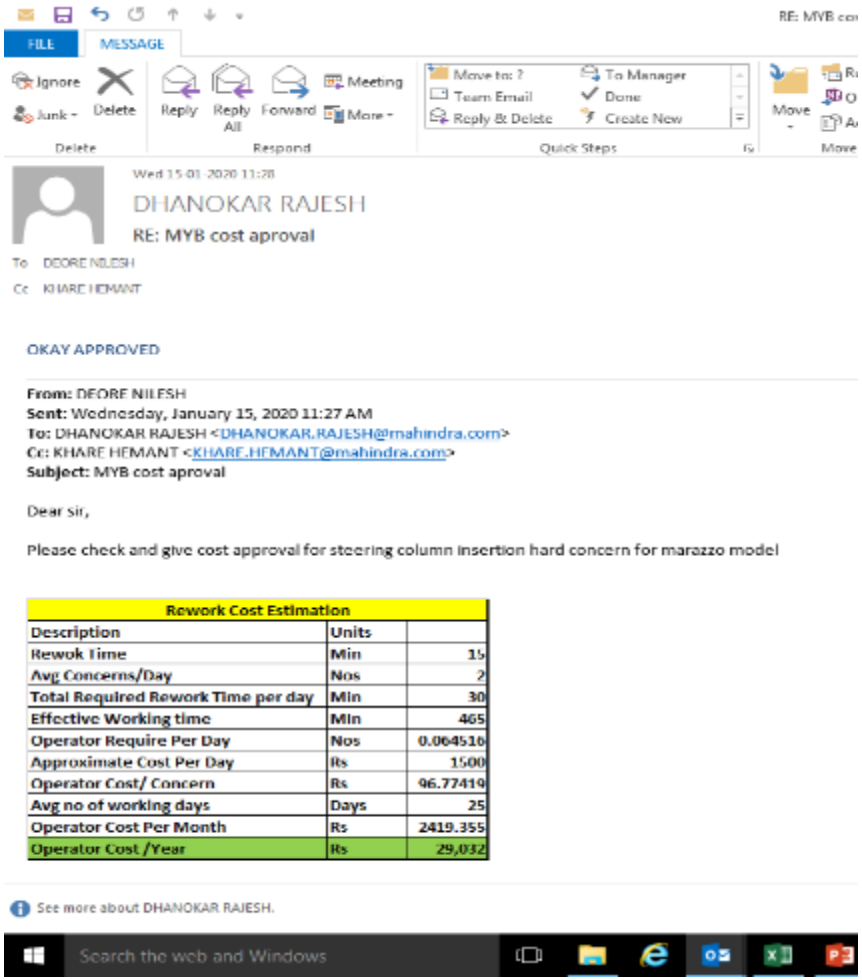
7.3 Tangible & Intangible benefits:

Tangible Benefits -

1) Chronic concern is eliminated.
2) COPQ (Material cost+ Labour cost+ Rework cost) =(Apprx. Rs.29,032 Per annum)is Saved

In tangible Benefits

1) Knowledge of Worker is Improved.
2) Improved Customer Satisfaction.



RE: MYB co

FILE MESSAGE

Ignore Junk - Delete | Reply Reply All Forward | Meeting More - | Move to: ? Team Email Reply & Delete | To Manager Done Create New | Move

Delete Respond Quick Steps Move

Wed 15-01-2020 11:28

DHANOKAR RAJESH

RE: MYB cost aproval

To DEORE NILESH

Cc KHARE HEMANT

OKAY APPROVED

From: DEORE NILESH
Sent: Wednesday, January 15, 2020 11:27 AM
To: DHANOKAR RAJESH <DHANOKAR.RAJESH@mahindra.com>
Cc: KHARE HEMANT <KHARE.HEMANT@mahindra.com>
Subject: MYB cost aproval

Dear sir,

Please check and give cost approval for steering column insertion hard concern for marazzo model

Rework Cost Estimation		
Description	Units	
Rewok Time	Min	15
Avg Concerns/Day	Nos	2
Total Required Rework Time per day	Min	30
Effective Working time	Min	465
Operator Require Per Day	Nos	0.064516
Approximate Cost Per Day	Rs	1500
Operator Cost/ Concern	Rs	96.77419
Avg no of working days	Days	25
Operator Cost Per Month	Rs	2419.355
Operator Cost /Year	Rs	29,032

See more about DHANOKAR RAJESH.

Search the web and Windows

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

SN	Document	Target date	Responsibility	Status
1	PCRN	27.10.2019	Sayali more	Closed
2	PFMEA updation.	27.10.2019	Suhas Aher	Closed
3	SOS Updation.	15.7.2019	Hemant khare	Closed

8.2 Training:

SN	Training Identified	Target date	Responsibility	Status
1.	On job training given to Body drop group and Roll down under pit Associate as per new PFC and SOS.	24.9.19	Hemant Khare	Closed

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

ENGINEERING NOTICE

DISPLAY PCRN

PCR...:* PCRN/A003/STCF/20/14553 | Use of tool for I shaft position setting | Planned closing date:* 27.10.2019 | Actual closing date:

A.Form

PU / Area:* STCF | Cell:* FAI | Date: 25.09.2019
Models Affected: Marazzo
Opn No./Mc No.: | Opn Description: Use of tool for I shaft position setting

Process Change Trigger

- [ ] Process Parameters
- [ ] Layout Change
- [ ] Production from New M/c
- [ ] Poka Yoka Introduction
- [ ] Inspection & Test Method
- [ ] Reconditioned M/c/ Tooling
- [ ] Method
- [ ] Use of alternate material
- [ ] 25% prodn. Ramp UP than rate quoted in present PPAP
- [ ] Transferred M/c
- [x] Any Other (Kaizens/ Improvement Projects/ Ingenious Ideas etc.)
- [ ] Tools
- [ ] Product produced after the tooling has been inactive for more than 12 month
- [ ] Change of Vendor for Part, Material, Paint etc
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Use of tool for I shaft position setting

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Estimated Cost Impact: ○ Saving ○ Increase ◉ Others

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- [ ] Conversion

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A.

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Justification:

MPPAP to be Followed:* ◉ Yes ○ No | SHE approval required ◉ Yes ○ No

Approval is necessary from (if affected)

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Approval Status

8. Standardization

8.1 Updation of documents: e.g. DFMEA, PFMEA, DCP, SOP, SOS, drawing.

PFMEA

Plant : Nashik			Model :Marazzo													Cross Functional Team			
Part No/s.:			Part Name/ Process:													Suhas Aher	Datta Jondhale		
Latest Change Letter:			Optn No.:0000064													Vikrant Ghule	Santosh Jadhav		
PU/ Shop :Scorpio TCF			Optn Desptn: UNLOADING CHASSIS ASSEMBLY ON FINAL LINE													Sayali More			
Module/ Cell :Final			M/C Name & No.:																
Process Function	**Requirements**	**Potential Failure Mode**	**Potential Effect(s) of Failures**	**sev**	**class**	**Potential Cause(s)/Mechanism(s) of Failures**	**Current Process Controls Prevention**	**Occur**	**Current Process Controls Detection**	**Detect**	**RPN**	**Recommended Action(s)**	**Responsibility & Target Completion Date**	**Action Results**					
														Action Taken	**Sev**	**Occ**	**Det**	**RPN**	
UNLOAD CHASSIS ON SLAT CONVEYOR .[TOGGLES PROVIDED ON SLAT CONVEYOR FOR LOCATING IN X DIRECTION].	COLLECTION OF PENDANT SWITCH & PRESS DOWN BUTTON OF SWITCH	CHASSIS NOT DROPED AT PROPER LOCATION ON SLAT CONVEYOR.	**Mfg. & Assy:** DIFFICULTY FOR FURTHER ASSEMBLY OF CHASSIS & BODY. **End user:** Nil	3		1)NO PROPER GUIDE FOR DROPPING CHASSIS. 2) TYRES NOT POSITIONED	1)TOGGLES PROVIDED ON SLAT CONVEYOR FOR LOCATION IN X DIRECTION 2) TWO GUIDES FOR FRT.AND RR WHEEL POSTIONING ON CONVEYOR IN Y DIRECTION	3	VEHICLE MISALIGNMENT WILL BE OBSERVED AT SAME STAGE	3	27				3				
CONNECT THE VACUUM HOSE[A] INTO MODULATOR [P] AND CONNECT THE CLIP [M] INTO ENGINE ASSEMBLY AS SHOWN IN FIGURE.	COLLECTION OF VACUUM HOSE	IMPROPER ROUTING OF VACUUM HOSE	**Mfg. & Assy:** PROCESS DELAY **End user:** Nil	4		OPERATOR MISTAKE	TRAINING TO OPERATOR & ROUTING IS MENTIONED IN SOP	2	AT BODY DROP STAGE	7	56	ASSOCIATE TRAINING	PRODUCTION TEAM		4				
POSOTION OF I SHAFT	MATCH I SHAFT MARKING WITH STEARING GEAR MARKING	INCORRECT POSITION OF I SHAFT	**Mfg. & Assy:** STEARING COLUMN INTERCONNECTION WILL NOT HAPPEN **End user:** Nil	4		OPERATOR MISTAKEDUEB TO EXCESS FORCE REQUIRED TO DO CORRECT POSITION	TRAINING TO OPERATOR & ROUTING IS MENTIONED IN SOP	2	AT BODY DROP STAGE	2	16	HAND TOOL TO BE PROVIDE FOR EASY POSITIONING OF I SHAFT	Pravin patil	Action completed					

Rev No.	***Rev Date***	***Revision Summary***	***Reason for Revision***
01	18-07-2017	New Release	
02	21.02.2018	Change in OCC	Six Monthly Review
03	07.08.2018	NO Change	Six Monthly Review
04	12.02.2019	NO Change	Six Monthly Review
05	05.08.2019	NO Change	Six Monthly Review
6	27.10.2019	Updated for introduction of tool for I shaft postion setting	PCRN/A003/STCF/2014553

8. Standardization

8.1 Updation of documents: e.g. PFMEA, DCP, SOP, SOS,

SOS

POS N°: SOS ...../.....

Standard Operation Sheet (continued)

Page ...../.....

(ANALYSIS)

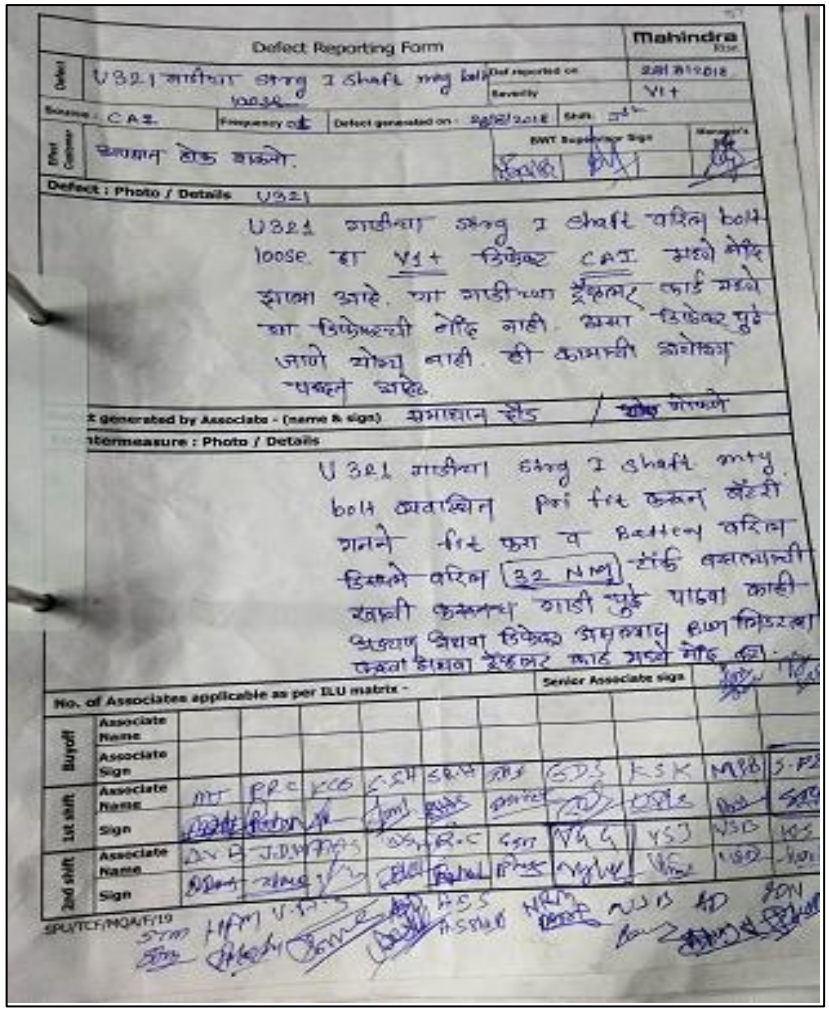
N°	Analysis of the operation	Time	Main step	Key point	Reason for the key point	Explanatory drawing. Operating and other rules.
14	Rotate with RH steering gear pinion upto matching white marking on steering rack			2 Upto match white marking		Steering gear rotating fixture
15	Remove with RH steering gear fixture & place it on its location					

What is forbidden and why (Explanation of possible problems or faults)

How to deal with anomalies (Explanatory items or notes)

8. Standardization

8.2 Training


Defect Reporting Form

Mahindra

Defect: U321 गाडीचा Stng I Shaft मध्ये bolt loose

Defect reported on: 28/8/2018

Severity: V1+

Business: CAI

Frequency of: 1

Defect generated on: 28/8/2018

Shift: 3rd

Effect Customer: समाधान होऊ शकतो.

SWT Supervisor Sign

Manager's

Defect : Photo / Details U321

U321 गाडीचा Stng I Shaft वरील bolt loose हा V1+ डिफेक्ट CAI मध्ये नोंद झाला आहे. या गाडीच्या ट्रॅकर कार्ड मध्ये या डिफेक्टची नोंद नाही. असा डिफेक्ट पुढे जाणे योग्य नाही. ही कामाची क्वालिटी राखत आहे.

Defect generated by Associate - (name & sign) समाधान होऊ / नोंद नोंदवले

Intermeasure : Photo / Details

U321 गाडीचा Stng I Shaft मध्ये bolt असावयाचा fix करून खोडरी गनने fix करा व Battery वरील डिस्कने वरील [32 Nm] टॉर्क बसलाच्ची खात्री करूनच गाडी पुढे पाठवा काही अडचण असेल डिफेक्ट असल्यास BOM किंवा करता डायरा ट्रॅकर कार्ड मध्ये नोंद करा.

No. of Associates applicable as per ILU matrix -

Senior Associate sign

Buyoff	Associate Name										
	Associate Sign										
1st shift	Associate Name	MT	PRC	KCG	CSH	SRH	[illegible]	GDS	KSK	MPB	S.PS
	Sign	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
2nd shift	Associate Name	DV	R	J.D.H	[illegible]	DS	R.C	GSD	MHG	YSJ	NSD	KKS
	Sign	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

SPU/TCF/NQA/F/19

Operator communication and On job Training given to operators :

Resp Hemant Khare :

Sep 19 : Closed

8. Standardization

8.3 Additional Audit / Check

6	Check point added in Under Pit check sheet.	17.10.2018	Hemant Khare	Closed

U321 Under Pit BUY-OFF - MARKING STANDARD								
					Rev no:- 00			Revision on -
								17.10.2018
Sr. No.	SC CC	Description	What to Check	Sev.	How to Check	Why To check	How to Mark	Type of Mark
6	CC	I Shaft	1.Bolt- No loose fitment (1 Nos.)	V1+	1.Visual and Marking.	1.Fatal accident will happen 2.Rattling Noise.		Yellow Paint Mark

9. Summaries & Plan future actions

9.1 Conclusion:

I shaft column insertion hard eliminated .

Handy fixture provided for positioning of power steering rack .

9.2 Horizontal Deployment:

System given for implementation at Nashik Bolero Line, Haridwar, Zahirabad & Chakan plants

9.3 Lessons Learnt:

Things Gone Wrong :

Associate complexity not consider with increasing number of Models on same line.

Things Gone Right : .

Using the HEP principles we are able to eliminate the all failure modes & able to make the operation error proof.

Thank You...